

Predicting Shipment Delays

Summary of Analysis & Strategic Recommendations

Rami Albatal

Tasks



Analyse historical data and estimate on-time delivery



Predict the likelihood of shipment delays using historical data

Exploratory Analysis

GPS Data



- **Some carriers have opted out of tracking, providing only one GPS record per shipment.**
- **Data is often incomplete, with tracking stopping unexpectedly during some shipments.**
- **Carriers sometimes move closer to the destination, then away, possibly forgetting to stop tracking.**
- **There is significant variation in GPS signal frequency:**
 - **25th percentile: 0.89 signals per 5 minutes**
 - **Median: 2.42 signals per 5 minutes**
 - **75th percentile: 3.50 signals per 5 minutes.**
- **The last GPS record may not be the closest to the destination.**

Exploratory Analysis



Shipment Bookings

- **Delivery and Collection times**
 - In some cases, the delivery schedule window appears to be earlier than the collection window.
 - In some cases, the latest time in a window (collection or delivery) is before the earliest time.
- **Postcodes**
 - Some entries in the shipment bookings are missing postcodes of collection or delivery.
 - External API (<https://api.opencagedata.co>) was used in order to fill the missing values by using the GPS coordinate.

First Task: Percentage of On-Time Deliveries

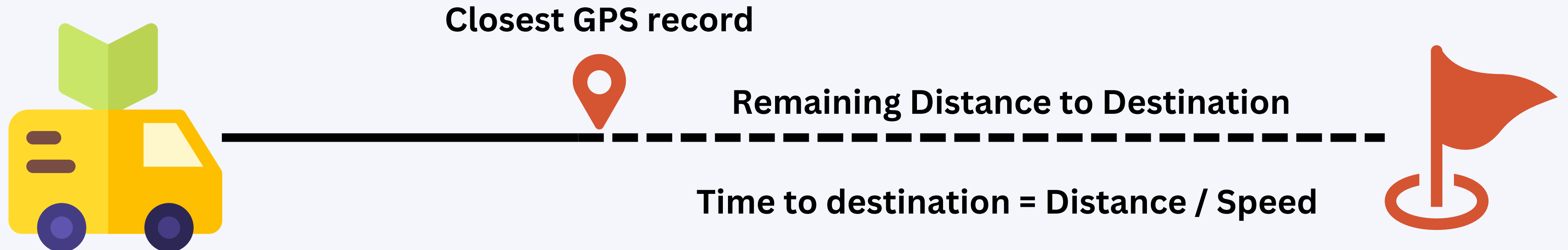
Assumptions

- **One GPS record indicates opted-out tracking.**
- **GPS tracking may continue after delivery, and may stop unexpectedly.**
- **Successful delivery occurs within 0.5 miles of the destination.**
- **Delivery time estimates improve as distance decreases & GPS frequency increase:**
 - **Certain: ≤ 0.5 miles from destination**
 - **High Confidence: 0.5 to 50 miles away & GPS frequency $\geq$ Median**
 - **Moderate Confidence: 0.5 to 50 miles away & GPS frequency $<$ Median**
 - **Moderate Confidence: > 50 miles away & GPS frequency $\geq$ Median**
 - **Low Confidence: > 50 miles away & GPS frequency $<$ Median**

First Task: Percentage of On-Time Deliveries

Approach

- Identify the closest GPS record to the destination for each shipment.
- Calculate the movement speed at that GPS record and the distance remaining.
- Estimate the time needed to reach the destination using speed and distance.
- Check if the estimated arrival time fits within the delivery window.



First Task: Percentage of On-Time Deliveries

Results

- **Between October 1st and December 31st, 2023**
- **Total shipments analysed: 3,237**
- **On-time arrival estimate: 68.89% (2,230 shipments)**
 - **1,648 certain, 177 high confidence, 249 moderate confidence, 156 low confidence**
- **Late arrival estimate: 26.32% (852 shipments)**
- **Percentage opting out of GPS tracking: 4.79% (155 shipments)**

Second Task: Likelihood of delay in new data

Pre-processing

- **Implement data pre-processing based on the insights from the previous task:**
 - **Rectify timing discrepancies in collection and delivery windows**
 - **Complete any missing postcode entries**
 - **Verify that latitude and longitude values are within a sensible range for shipments in the UK and Ireland**

Second Task: Likelihood of delay in new data

Features

- **Identify and extract key features that are thought to influence the prediction of whether a shipment will arrive on time or not.**
 - a. **Location-related features:**
 - i. **The total distance between the starting and ending locations**
 - ii. **The outer segments of the collection and delivery postcodes (to represent the area)**
 - b. **Time-related features:**
 - i. **The day of the week and the hour of the collection and delivery time frames**
 - ii. **The duration of the collection and delivery time windows**
 - c. **Additional features: Carrier ID, Vehicle Size, Vehicle Composition**

Second Task: Likelihood of delay in new data

Training dataset

- **Pre-process and extract features from historical data (Shipment_bookings.csv) for training.**
- **Ground truth is the on-time delivery estimation (calculated in the first task); True indicates positive samples, False indicates negative samples.**
- **The training data is imbalanced, with positive samples being 2.6 times more than negative. A technique called *oversampling* was used to enhance negative sample representation.**

Second Task: Likelihood of delay in new data

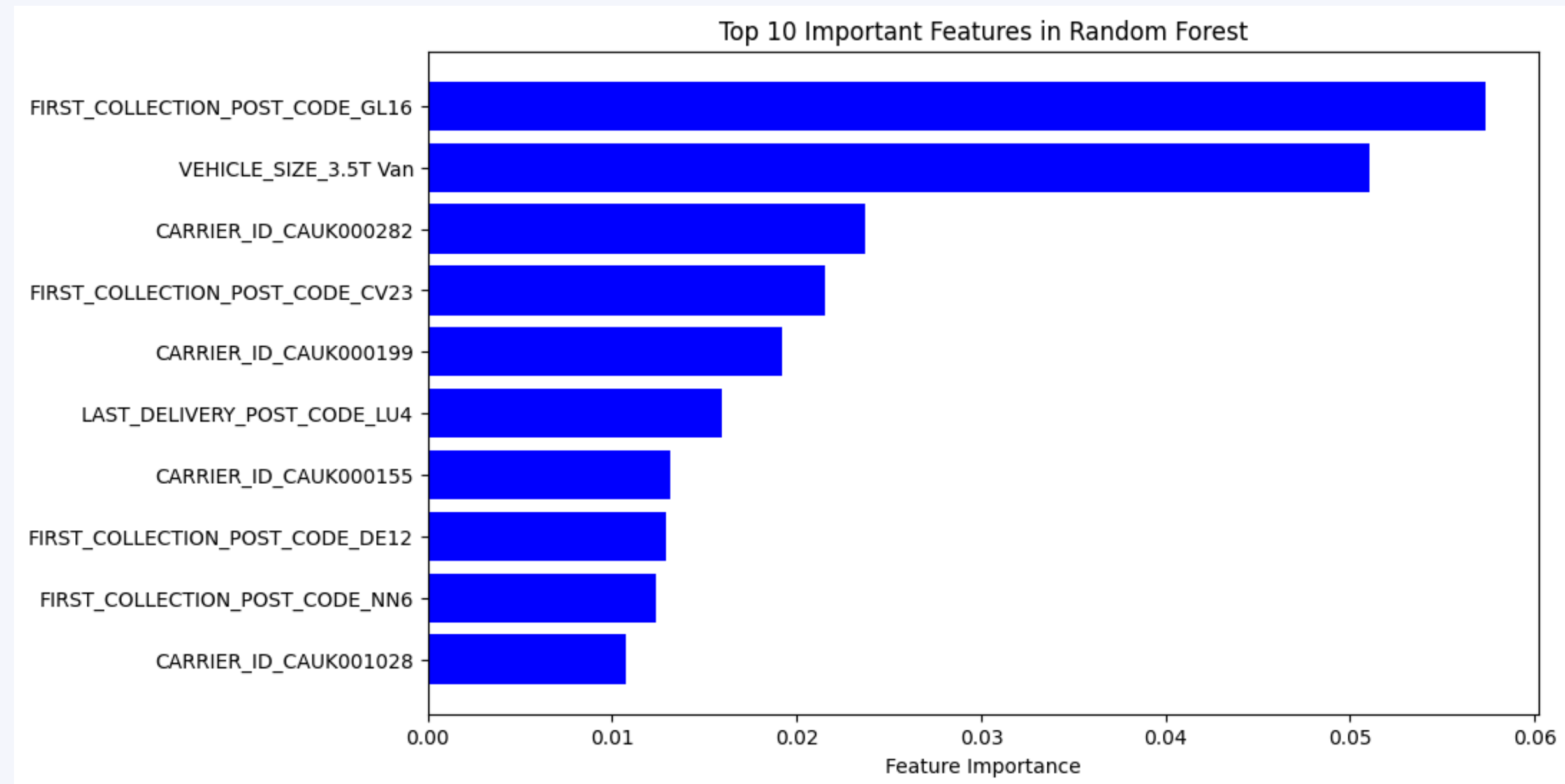
Training an AI Model to Predict Delay Likelihood

- **Three predictive machine learning models were developed and evaluated:**
 - **Logistic Regression:** A baseline model for binary classification, achieving an accuracy of **67.75%**
 - **Random Forest:** A model that utilizes multiple decision trees, with an accuracy of **70.66%**
 - **XGBoost:** An advanced model featuring additive boosting algorithm, reaching an accuracy of **71.80%**
- **Our conclusion is that XGBoost is the most effective model to implement.**

Second Task: Likelihood of delay in new data

Features Importance

Top features for estimating delivery time include location data (postcodes), specific carriers, and vehicle types, which help identify causes of shipment delays.



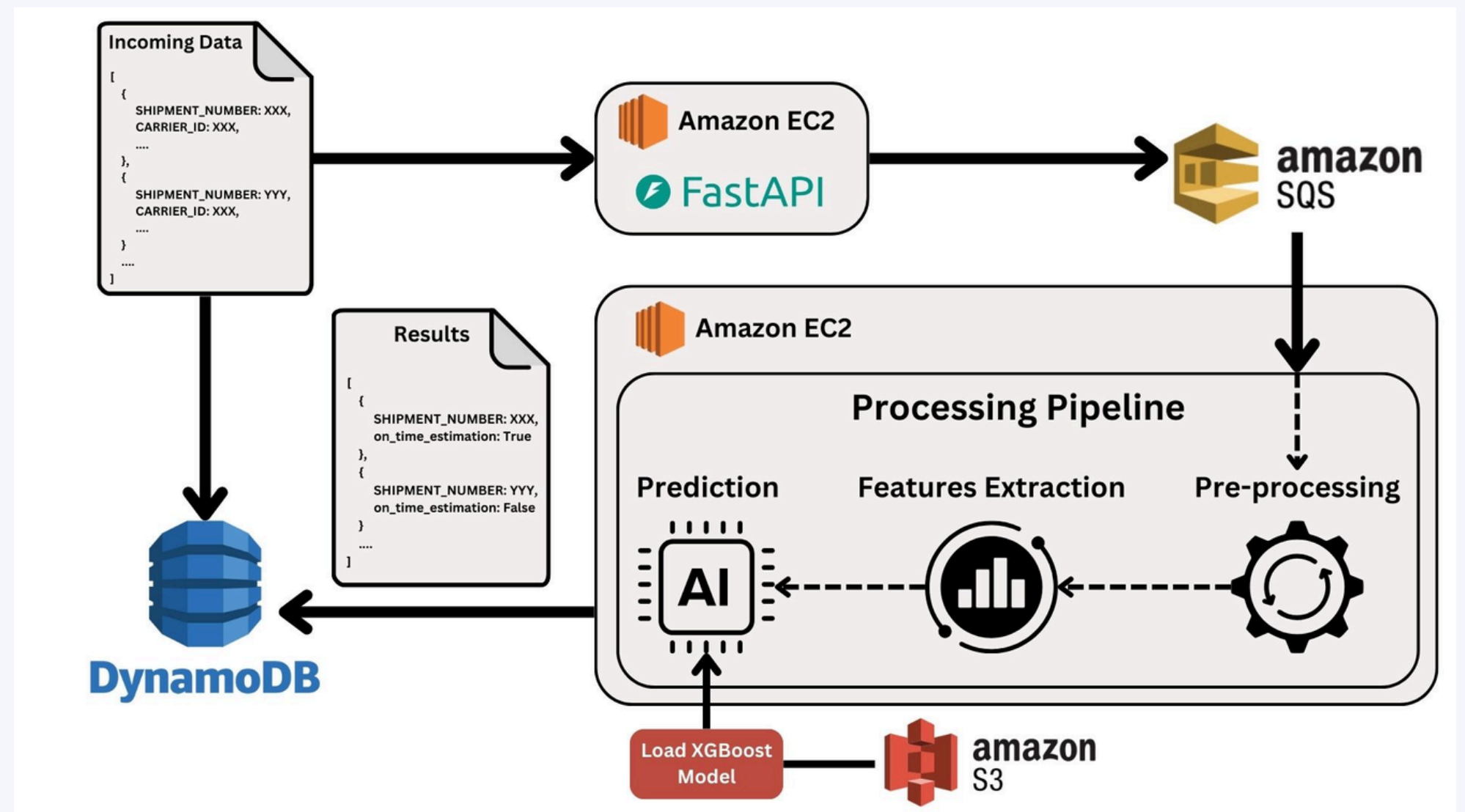
Second Task: Likelihood of delay in new data

Testing

- Pre-process and extract features from New_bookings.csv for testing.
- Use the XGBoost model to predict on-time delivery probability for the testing dataset.
- Calculate delay likelihood as $1 - \text{probability of on-time delivery}$.
- Results:
 - Shipments likely to be delayed: 265
 - Shipments likely to arrive on time: 735
 - Delay likelihood in new data: 26.5%

Technical design for deploying the prediction model through an online endpoint

- AWS Cloud
- API processes shipment requests
- Requests added to a queue
- Processing pipeline reads from the queue and performs pre-processing, feature extraction, and prediction
- Results stored in a database
- Aspects to include: Scalability, Availability, Monitoring, Logging & Security



Thank you

Contact

Rami Albatal

Data Scientist

 ramibatal@gmail.com